

PLANETA FP

INTELIGENCIA ARTIFICIAL Y BIG DATA

GIT Y GITHUB

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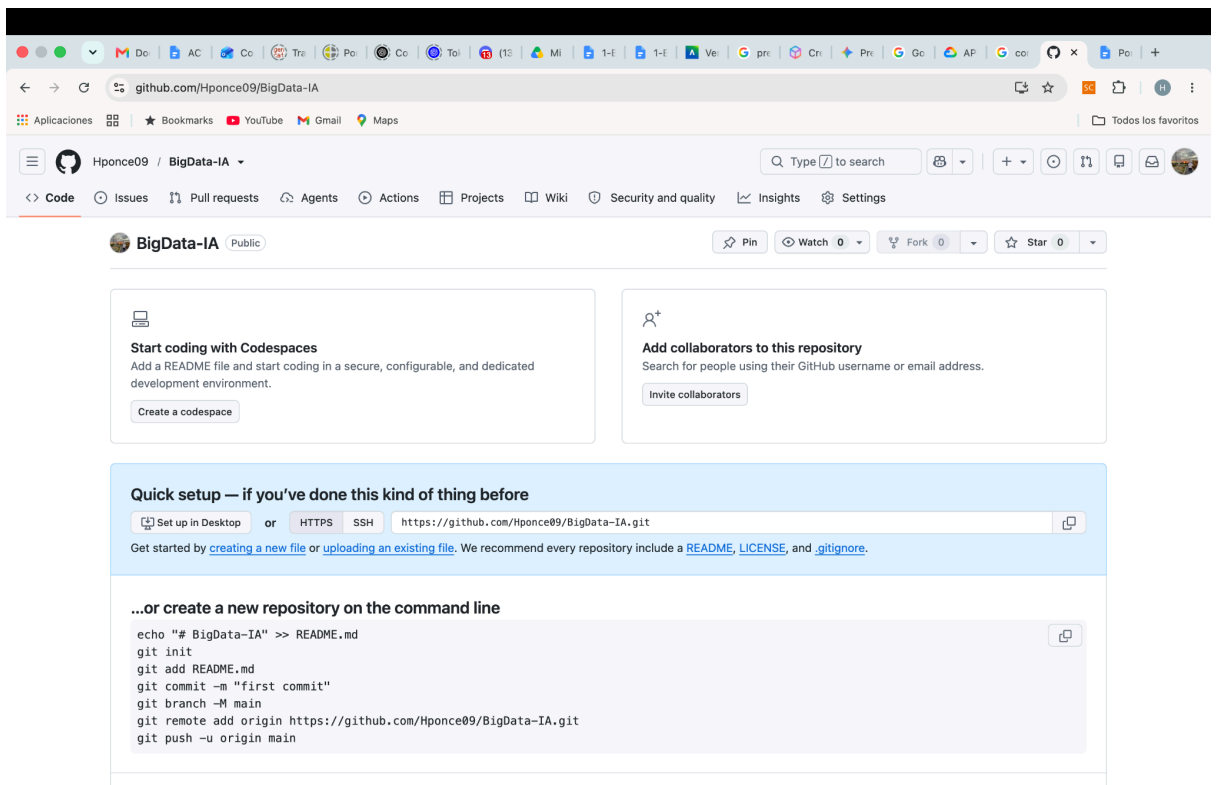
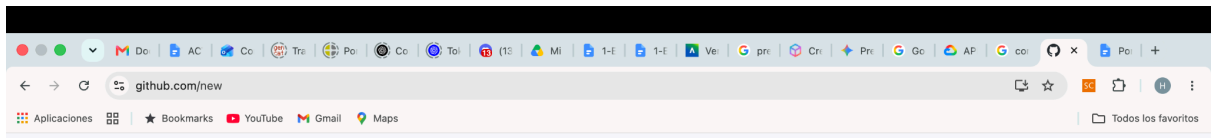
1. Cuenta GitHub

The screenshot shows the GitHub profile page for user Hponce09. The browser address bar displays 'github.com/Hponce09'. The profile header includes the username 'Hponce09', a search bar, and navigation links for Overview, Repositories (9), Projects, Packages, and Stars. The profile picture is a circular image of a person walking on a path. Below the profile picture are links for 'Edit profile', 'Achievements' (showing two gold medals), and 'Organizations' (showing one organization). The 'Pinned' section features three repositories: 'JohanSebastianPZ/Activity_Booking_Plugin' (Public, PHP), '2526-daw2-pf-c2-tagprice' (Public, PHP), and 'Activity_Booking_Plugin_RM' (Public). The '354 contributions in the last year' section displays a grid of green squares representing contributions across months from September to September. The 'Contribution activity' section shows a timeline for September 2026, indicating 'Created 1 repository' with the link 'Hponce09/clase-big-data' and a TypeScript icon.

inicio de sesión personal GitHub

2. Repositorio Publico

Creación del repositorio remoto en github, especificando que este es público y asignándole un nombre al repositorio



3. Conexión SSH

Generación de la clave SSH en local

```
andresponce@Address-MacBook-Air planetafp % ssh-keygen -t ed25519 -C "harold.andres.ponce@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/andresponce/.ssh/id_ed25519):
Enter passphrase for "/Users/andresponce/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /Users/andresponce/.ssh/id_ed25519
Your public key has been saved in /Users/andresponce/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:b1PfN2YsaPCJXUqUv3yoW4Qblrrre24qlNwTkong6Gng harold.andres.ponce@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|
|   .   .
|  o   o
| o . o . +
| o o S . * . +
| . . . .0oX.+
|..E . .o=X *.o|
|... .B++ = o|
|.. .=.*=.
+-----[SHA256]-----+
```

Añadir clave a mi cuenta personal github

The screenshot shows the GitHub 'Add new SSH Key' page. The user's profile 'Hponce09 (Hponce09)' is visible at the top. The sidebar on the left lists various settings and navigation options. The main content area is titled 'Add new SSH Key' and contains a form with the following fields:

- Title:** A text input field containing 'Mi MacBook'.
- Key type:** A dropdown menu set to 'Authentication Key'.
- Key:** A large text area containing the SSH key: `ssh-rsa, 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'`

A green 'Add SSH key' button is located at the bottom of the form.

prueba de conexión SSH

```
[andresponce@Andress-MacBook-Air planetafp % eval "$(ssh-agent -s)"  
  
Agent pid 31905  
[andresponce@Andress-MacBook-Air planetafp % ssh-add --apple-use-keychain ~/.ssh/id_ed25519  
  
Enter passphrase for /Users/andresponce/.ssh/id_ed25519:  
Identity added: /Users/andresponce/.ssh/id_ed25519 (harold.andres.ponce@gmail.com)  
andresponce@Andress-MacBook-Air planetafp % ssh -T git@github.com  
  
Hi Hponce09! You've successfully authenticated, but GitHub does not provide shell access.  
andresponce@Andress-MacBook-Air planetafp % █
```

4. Clonar repositorio

Clonación del repositorio remoto en local para su debida vinculación

```
[andresponce@Andress-MacBook-Air planetafp % git clone git@github.com:Hponce09/BigData-IA.git  
Cloning into 'BigData-IA'...  
warning: You appear to have cloned an empty repository.  
[andresponce@Andress-MacBook-Air planetafp % ls  
BigData-IA  
[andresponce@Andress-MacBook-Air planetafp % cd BigData-IA  
andresponce@Andress-MacBook-Air BigData-IA % █
```

5. Estructura de carpetas

Creación de las diferentes carpetas de las asignaturas con las subcarpetas de trabajo

machine learning

```
andresponce@Andress-MacBook-Air BigData-IA % mkdir -p Machine_Learning/teoria Machine_Learning/ejercicios
andresponce@Andress-MacBook-Air BigData-IA % ls
Machine_Learning
andresponce@Andress-MacBook-Air BigData-IA % cd Machine_Learning
andresponce@Andress-MacBook-Air Machine_Learning % ls
ejercicios      teoria
andresponce@Andress-MacBook-Air Machine_Learning % touch Machine_Learning/teoria/.gitkeep Machine_Learning/ejercicios/.gitkeep
touch: Machine_Learning/teoria/.gitkeep: No such file or directory
touch: Machine_Learning/ejercicios/.gitkeep: No such file or directory
andresponce@Andress-MacBook-Air Machine_Learning % ls
ejercicios      teoria
andresponce@Andress-MacBook-Air Machine_Learning % cd ejercccios
cd: no such file or directory: ejercccios
andresponce@Andress-MacBook-Air Machine_Learning % cd ejercicios
andresponce@Andress-MacBook-Air ejercicios % touch .gitkeep
andresponce@Andress-MacBook-Air ejercicios % ls
andresponce@Andress-MacBook-Air ejercicios % ls
andresponce@Andress-MacBook-Air ejercicios % touch .gitkeep
andresponce@Andress-MacBook-Air ejercicios % ls -a
.
..
.gitkeep
andresponce@Andress-MacBook-Air ejercicios % cd ..
andresponce@Andress-MacBook-Air Machine_Learning % ls
ejercicios      teoria
andresponce@Andress-MacBook-Air Machine_Learning % cd teoria
andresponce@Andress-MacBook-Air teoria % touch .gitkeep
andresponce@Andress-MacBook-Air teoria % ls -a
.
..
.gitkeep
```

Programación

```
andresponce@Andress-MacBook-Air BigData-IA % mkdir -p Programacion/teoria Programacion/ejercicios
andresponce@Andress-MacBook-Air BigData-IA % ls
Machine_Learning      Programacion
andresponce@Andress-MacBook-Air BigData-IA % cd Progrmacion
cd: no such file or directory: Progrmacion
andresponce@Andress-MacBook-Air BigData-IA % cd Programacion
andresponce@Andress-MacBook-Air Programacion % touch teoria/.gitkeep ejercicios/.gitkeep
```

Proyectos

```
andresponce@Andress-MacBook-Air BigData-IA % mkdir -p Proyectos/teoria Proyectos/ejercicios
andresponce@Andress-MacBook-Air BigData-IA % ls
Machine_Learning      Programacion      Proyectos
andresponce@Andress-MacBook-Air BigData-IA % touch Proyectos/teoria/.gitkeep Proyectos/ejercicios/.gitkeep
```

Models IA

```
andresponce@Andress-MacBook-Air BigData-IA % ls
Machine_Learning      Programacion      Proyectos
andresponce@Andress-MacBook-Air BigData-IA % mkdir -p Models_IA/teoria Models_IA/ejercicios
andresponce@Andress-MacBook-Air BigData-IA % touch Models_IA/teoria/.gitkeep Models_IA/ejercicios/.gitkeep
andresponce@Andress-MacBook-Air BigData-IA % ls
Machine_Learning      Models_IA      Programacion      Proyectos
andresponce@Andress-MacBook-Air BigData-IA % █
```

6. Commit por cada asignatura

Se realizaron los correspondiente “git add” y “git -m commit” para cada uno de los cambios de los diferentes archivos

machine learning

```
[andresponce@Andress-MacBook-Air teoria % git status]
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   ../ejercicios/.gitkeep
    new file:   .gitkeep

[andresponce@Andress-MacBook-Air teoria % git commit -m "docs(machine-learning):crear estructura de carpetas"]
[main (root-commit) 2da7a93] docs(machine-learning):crear estructura de carpetas
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Machine_Learning/ejercicios/.gitkeep
 create mode 100644 Machine_Learning/teoria/.gitkeep
[andresponce@Andress-MacBook-Air teoria % git push]
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 349 bytes | 349.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hponce09/BigData-IA.git
 * [new branch]      main -> main
andresponce@Andress-MacBook-Air teoria %
```

Programación

```
[andresponce@Andress-MacBook-Air BigData-IA % git add Programacion]
[andresponce@Andress-MacBook-Air BigData-IA % git status]
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Programacion/ejercicios/.gitkeep
    new file:   Programacion/teoria/.gitkeep

[andresponce@Andress-MacBook-Air BigData-IA % git commit -m "docs(programacion):crear estructura de carpetas"]
[main 12ef3a6] docs(programacion):crear estructura de carpetas
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Programacion/ejercicios/.gitkeep
 create mode 100644 Programacion/teoria/.gitkeep
[andresponce@Andress-MacBook-Air BigData-IA % git push]
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 281 bytes | 281.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hponce09/BigData-IA.git
 2da7a93..12ef3a6  main -> main
andresponce@Andress-MacBook-Air BigData-IA %
```

Proyectos

```
andresponce@Andress-MacBook-Air BigData-IA % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Proyectos/

nothing added to commit but untracked files present (use "git add" to track)
andresponce@Andress-MacBook-Air BigData-IA % git add Proyectos
andresponce@Andress-MacBook-Air BigData-IA % git commit -m "docs(proyectos):crear estructura de carpetas"
[main 854d025] docs(proyectos):crear estructura de carpetas
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Proyectos/ejercicios/.gitkeep
 create mode 100644 Proyectos/teoria/.gitkeep
andresponce@Andress-MacBook-Air BigData-IA % git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 280 bytes | 280.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hponce09/BigData-IA.git
 12ef3a6..854d025  main -> main
```

Models IA

```
andresponce@Andress-MacBook-Air BigData-IA % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Models_IA/

nothing added to commit but untracked files present (use "git add" to track)
andresponce@Andress-MacBook-Air BigData-IA % git add Models_IA
andresponce@Andress-MacBook-Air BigData-IA % git commit -m "docs(models_ia):crear estructura de carpetas"
[main 1cfdcab] docs(models_ia):crear estructura de carpetas
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Models_IA/ejercicios/.gitkeep
 create mode 100644 Models_IA/teoria/.gitkeep
andresponce@Andress-MacBook-Air BigData-IA % git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 264 bytes | 264.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hponce09/BigData-IA.git
 854d025..1cfdcab  main -> main
andresponce@Andress-MacBook-Air BigData-IA %
```

7. Subir Documentación

Subir las carpetas a Github y las asignación de la documentación a las correspondientes carpetas según su asignatura.

The screenshot shows the GitHub repository page for 'BigData-IA' by user 'Hponce09'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows 16 commits, with the latest commit being 'docs(programacion):add ejercicios del 9-10 programacion' by Hponce09, dated yesterday. The repository structure is as follows:

File/Folder	Commit Message	Commit Date
Machine_Learning	docs(models_ia):add apuntes que es ML Machine_Learning	last week
Models_IA	docs(models_ia):add apuntes de modelos de IA Models_IA	last week
Programacion	docs(programacion):add ejercicios del 9-10 programacion	yesterday
Proyectos	docs(proyectos):add ejercicios github Proyectos	last week
.DS_Store	docs(programacion):add ejercicios del 1-5 y anexo de la te...	5 days ago

The screenshot shows the GitHub repository file explorer and code view. The file explorer on the left shows the repository structure, with the 'Programacion' folder expanded, showing 'ejercicios' and 'teoria' subfolders. The 'ejercicios' folder contains '1-Ejercicios.docx', '2-Ejercicios_DC.py', and '2-ejercicios_datos_compuesto...'. The 'teoria' folder contains '1-Tipos.docx', '2-DatosCompuestos_y_Condici...', and '\$DatosCompuestos_y_Condici...'. The code view on the right shows the content of '2-Ejercicios_DC.py', which is a Python script for a grade control exercise.

```
1 """
2 Ejercicio 1. Listas: control de notas
3 - Crea una lista llamada notas con cinco calificaciones: 6, 8, 5, 9 y 7.
4 - Guarda en una variable primera_notas el primer elemento de la lista.
5 - Guarda en una variable ultima_notas el último elemento de la lista.
6 - Cambia la segunda nota de la lista por un 10.
7 - Añade una nueva nota, 8, al final de la lista.
8 - Guarda en una variable total_notas la cantidad de notas que hay en la lista.
9 - Muestra por consola la lista final, la primera nota, la última nota y el total de notas.
10
11 """
12
13 print("Solucion de ejercicio 1")
14
15 notas = [6,8,5,9,7]
16 primera_notas = notas[0]
17 #ultima_notas = notas[len(notas)-1]
18 notas[1]=10
19 notas.append(8)
20 ultima_notas = notas[len(notas)-1]
21 total_notas = len(notas)
22
23 print(notas)
24 print(primera_notas)
```

8 Historial

```
docs(machine_learning, teoria, estructura de carpetas)
[andresponce@Andress-MacBook-Air BigData-IA % git log --oneline --graph --decorate --all

* 5c9b578 (HEAD -> main, origin/main) docs(programacion):add ejercicios del 9-10 programacion
* a4665d9 docs(programacion):add ejercicios del 8 programacion
* bc13ef5 docs(programacion):add ejercicios del 6-7 programacion
* 23be0d2 docs(programacion):add ejercicios del 1-5 y anexo de la teoria de datos compuestos Programacion
* 25184c9 docs(programacion):add ejercicios datos compuestos Programacion
* 809f719 docs(proyectos):add ejercicios github Proyectos
* 331e73d docs(models_ia):add apuntes datos compuestos Programacion
* e47daa4 docs(models_ia):add apuntes que es ML Machine_Learning
* edacd91 docs(models_ia):add apuntes de modelos de IA Models_IA
* 6b716b5 docs(programacion):add apuntes de tipos de datos Programacion
* 2ba90c1 docs(programacion):add ejercicios Programacion
* 64d9724 docs(machine_learning):add ejercicios ML
* 1cfdca4 docs(models_ia):crear estructura de carpetas
* 854d025 docs(proyectos):crear estructura de carpetas
* 12ef3a6 docs(programacion):crear estructura de carpetas
* 2da7a93 docs(machine-learning):crear estructura de carpetas
andresponce@Andress-MacBook-Air BigData-IA %
```

9 Conclusión

En conclusión esta práctica nos muestra como git y github trabajan en conjunto, para llevar un registro de los cambios en el proyecto, además su funcionamiento tiene diferentes sistemas de seguridad para vincular los repositorios entre sí. En conclusión es una herramienta de trabajo, de la cual podemos sacar mucho provecho, pero también debemos conocerla más a profundidad, para sacar su máximo aprovechamiento a beneficio nuestro y entorno de trabajo.